Hardware Requirements Guide

Running Clawdbot + Seedbox + Media Server

Overview

This guide outlines the hardware specifications needed to run three services simultaneously on a single machine:

1. **Clawdbot** — Your AI assistant and automation platform
2. **Seedbox** — BitTorrent client for downloading/seeding
3. **Media Server** — Plex/Jellyfin for streaming media

Three Configuration Options

Option 1: Budget Build (~$300-500)

Good for: Clawdbot + light seedbox + media server for 1-2 users

Component	Specs	Notes
CPU	Intel i3-10100 / AMD Ryzen 3 3200G	4 cores, integrated graphics
RAM	8 GB DDR4	Single stick okay, dual-channel better
Storage (OS)	120 GB SSD	SATA or NVMe
Storage (Media)	2-4 TB HDD	5400 RPM minimum, 7200 RPM preferred
Network	Gigabit Ethernet	Must have stable internet
Power Supply	300W 80+ Bronze	Don’t cheap out here

Component	Specs	Notes
Case	Any mini-ITX or micro-ATX	Ensure good airflow

Estimated Power Draw: 30-60W idle, 80-120W under load

Option 2: Recommended Build (~$600-900)

Good for: Clawdbot + active seedbox + media server for 3-5 concurrent users

Component	Specs	Notes
CPU	Intel i5-12400 / AMD Ryzen 5 5600G	6 cores, excellent efficiency
RAM	16 GB DDR4 (2x8GB)	Dual-channel for better performance
Storage (OS/Apps)	256 GB NVMe SSD	Fast boot and app loading
Storage (Media)	4-8 TB HDD (or NAS)	Consider 2x drives for redundancy
Network	Gigabit Ethernet	Cat6 cables, gigabit router
Power Supply	450W 80+ Gold	Efficiency matters for 24/7 operation
Case	Fractal Design Node 304 / SilverStone	Hot-swappable bays ideal

Estimated Power Draw: 35-70W idle, 100-150W under load

Option 3: High-Performance Build (~$1200-1800)

Good for: Clawdbot + heavy seedbox (1000+ torrents) + 4K media server for 10+ users

Component	Specs	Notes
CPU	Intel i5-13500 / AMD Ryzen 7 5700G	8+ cores, hardware transcoding
RAM	32 GB DDR4 (2x16GB)	Future-proof, handles many

Component	Specs	Notes
		transcodes
Storage (OS/Apps)	500 GB NVMe SSD	Samsung 980 Pro or similar
Storage (Media)	8-16 TB (multiple drives)	Consider RAID/ZFS setup
Storage (Cache)	1 TB NVMe SSD	For seedbox cache/transcoding
Network	2.5GbE or 10GbE	If your internet supports it
Power Supply	550W 80+ Platinum	Maximum efficiency for 24/7
Case	Fractal Design Define 7 / SilverStone CS381	8+ drive bays, good cooling

Estimated Power Draw: 45-90W idle, 150-250W under load

Component Deep Dive

CPU Requirements

Minimum: - 2 cores / 4 threads - x86_64 architecture (Intel or AMD) - Supports hardware virtualization (for Docker/containers)

Recommended: - 4+ cores / 8+ threads - Intel Quick Sync or AMD VCE (for hardware transcoding) - Low TDP (65W or less) for efficiency

Why it matters: - Clawdbot: 1-2 cores typical usage - Seedbox: 1-2 cores during active downloads - Media Server: 1-4 cores per transcode (software) or minimal (hardware)

RAM Requirements

By Service: | Service | Minimum | Recommended | Heavy Use | |———|———|————-|———–| | Clawdbot | 1 GB | 2 GB | 4 GB | | Seedbox (qBittorrent) | 512 MB | 1-2 GB | 4 GB+ | | Media Server (Plex) | 1 GB | 2 GB | 4 GB+ | | OS Overhead | 1-2 GB | 2 GB | 4 GB | | **Total** | **4 GB** | **8 GB** | **16 GB** |

RAM Tips: - DDR4 is fine; DDR5 not necessary - ECC RAM not required (nice for ZFS though) - Leave room for growth

Storage Requirements

SSD (Operating System + Applications): - 120 GB minimum - 256 GB recommended (room for logs, temp files) - NVMe preferred for faster boot times

HDD (Media Storage): - Calculate: ~3-5 GB per hour of 1080p content - 4K content: ~15-25 GB per hour - Recommendation: Start with 4 TB, expand as needed

Seedbox Storage: - Active downloads: 100-500 GB - Seeding: Depends on ratio goals (1-4 TB typical)

Configuration Options:

```
Option A: Simple
├── 256 GB NVMe (OS + Apps + Clawdbot)
└── 4 TB HDD (Media + Seedbox)

Option B: Separate Cache
├── 256 GB NVMe (OS + Apps)
├── 1 TB NVMe (Seedbox cache + Transcoding)
└── 8 TB HDD (Media storage)

Option C: NAS/SAN
├── 256 GB NVMe (OS + Apps)
└── Network attached storage (Media + Seedbox)
```

Network Requirements

Internet: - Download: 50+ Mbps (media streaming) - Upload: 20+ Mbps (seedbox sharing, Clawdbot webhooks) - Unlimited data preferred

Local Network: - Gigabit ethernet minimum - Cat5e or Cat6 cables - Gigabit router/switch

Port Requirements: | Service | Ports | Direction | |———|——-|————–| | Clawdbot | 443, 80 | Inbound | | SSH | 22 | Inbound (restricted) | | Seedbox | 6881-6889 | Inbound/Outbound | | Plex | 32400 | Inbound | | Jellyfin | 8096 | Inbound |

Software Stack Recommendations

Operating System

Ubuntu Server 22.04 LTS (Recommended) - Stable, long-term support - Large community - Easy package management

Alternative: TrueNAS SCALE - Built-in ZFS - Docker support - Web-based management

Clawdbot Setup

```
# Install Node.js 20.x
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs

# Install Clawdbot
sudo npm install -g clawdbot

# Configure auto-start
sudo systemctl enable clawdbot
```

Seedbox Setup

Recommended: qBittorrent + VPN

```
# Install qBittorrent-nox (headless)
sudo apt install qbittorrent-nox

# Run as service
sudo systemctl enable qbittorrent-nox
```

Docker Alternative:

```
version: '3'
services:
  qbittorrent:
    image: linuxserver/qbittorrent
    container_name: qbittorrent
    environment:
      - PUID=1000
      - PGID=1000
    volumes:
      - ./config:/config
      - /mnt/media/downloads:/downloads
    ports:
      - 8080:8080
      - 6881:6881
    restart: unless-stopped
```

Media Server Setup

Plex Media Server:

```
# Download from plex.tv
wget https://downloads.plex.tv/plex-media-server-new/...
sudo dpkg -i plexmediaserver_*.deb
```

Jellyfin (Open Source Alternative):

```
# Install Jellyfin
curl -s https://repo.jellyfin.org/install-debuntu.sh | sudo bash
sudo apt install jellyfin
```

Power & Cost Analysis

Electricity Costs (24/7 Operation)

Build	Power Draw	Monthly Cost*	Annual Cost
Budget	50W average	$4.50	$54
Recommended	75W average	$6.75	$81
High-Performance	120W average	$10.80	$130

*Based on $0.12/kWh average North American rate

VPS vs Local Comparison (3-year TCO)

Cost	VPS ($20/mo)	Budget Local	Rec. Local	High-Perf Local
Hardware	$0	$400	$750	$1500
Electricity (3yr)	$0	$162	$243	$390
Internet	$0*	$0*	$0*	$0*
3-Year Total	**$720**	**$562**	**$993**	**$1890**

*Assumes you already pay for internet

Break-even point: ~18 months for budget build, ~28 months for recommended build

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Buying Guide

Where to Buy

New Components: - Amazon (convenience, returns) - Newegg (deals, bundles) - Micro Center (in-store deals, especially CPUs) - B&H Photo (no tax outside NY/NJ)

Used/Refurbished (Great for budget builds): - eBay (corporate off-lease desktops) - Facebook Marketplace (local deals) - r/homelabsales (Reddit community)

Recommended Pre-Built Options: - Dell OptiPlex 7050/7060 Micro (compact, quiet) - HP EliteDesk 800 G3 Mini - Lenovo ThinkCentre M720q

Sample Shopping List (Recommended Build)

Item	Model	Est. Price
CPU	Intel i5-12400	$180
Motherboard	B660 Micro-ATX	$100
RAM	16GB DDR4-3200	$50
SSD (OS)	256GB NVMe	$40
HDD (Media)	4TB WD Blue	$80
Case	Fractal Node 304	$100
PSU	450W 80+ Gold	$70
Total		**~$620**

Setup Tips

BIOS Settings

- Enable virtualization (VT-x/AMD-V) for Docker
- Set power profile to “Balanced” or “Power Saver”
- Disable unused features (audio if headless)

OS Optimization

```
# Reduce swappiness for SSD longevity
sudo sysctl vm.swappiness=10

# Enable automatic updates for security
sudo apt install unattended-upgrades

# Monitor temperatures
sudo apt install lm-sensors
sudo sensors-detect
sensors
```

Drive Management

```
# Check drive health
sudo smartctl -a /dev/sda

# Monitor disk usage
df -h
```

```
# Set up log rotation to prevent SSD wear
sudo nano /etc/logrotate.conf
```

Network Optimization

```
# Increase TCP buffer sizes
sudo sysctl -w net.core.rmem_max=134217728
sudo sysctl -w net.core.wmem_max=134217728

# Enable BBR congestion control
sudo nano /etc/sysctl.conf
# Add: net.ipv4.tcp_congestion_control=bbr
```

Troubleshooting Common Issues

High CPU Usage

- Check transcoding settings (use hardware acceleration)
- Limit concurrent seedbox connections
- Monitor with: `htop`, `iotop`

High RAM Usage

- Add swap file if needed: `sudo fallocate -l 4G /swapfile`
- Limit Plex transcoder threads
- Check for memory leaks: `ps aux --sort=-%mem | head`

Slow Network

- Test speed: `iperf3`, `speedtest-cli`
- Check cable quality
- Verify router QoS settings

Storage Full

- Set up automated cleanup scripts
- Use `ncdu` to find large files
- Implement retention policies for seedbox

Expansion Path

When you need more:

1. **More Storage:** Add external USB drives or NAS

2. **More RAM:** Easy upgrade on most motherboards
3. **Better Network:** Add 2.5GbE or 10GbE NIC
4. **Redundancy:** Add second machine for failover

Quick Reference

Minimum Viable Specs

- CPU: 4 threads
- RAM: 8 GB
- Storage: 120 GB SSD + 2 TB HDD
- Network: Gigabit ethernet

Recommended Specs

- CPU: 6 cores + iGPU
- RAM: 16 GB
- Storage: 256 GB NVMe + 4 TB HDD
- Network: Gigabit ethernet, good upload speed

Check Before Buying

- ☐ CPU supports Quick Sync / VCE?
- ☐ Motherboard has enough SATA ports?
- ☐ Case has room for all drives?
- ☐ Power supply sufficient wattage?
- ☐ Network card on motherboard?

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